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Project Synopsis (2025-26) - Sem VII

CredMill - an AI Credit Risk Assesser

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Sustainable Development Goals: Decent Work and Economic Growth

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Abstract

This project proposes CredMill, a resource-efficient, explainable, and fair AI-driven credit risk assessment system designed for small-to-medium-scale operational environments. Traditional manual credit evaluations are slow, prone to human inconsistency, and often fail to capture complex, nonlinear patterns in borrower data. While AI-based solutions have demonstrated improvements in predictive accuracy, they frequently function as opaque "black boxes," demand costly computational resources or proprietary APIs, and overlook the need for fairness auditing and bias mitigation. CredMill addresses these limitations by integrating lightweight yet high-performing machine learning models—particularly tree-based ensemble methods—with interpretability frameworks such as SHAP and LIME to ensure transparency at both global and individual decision levels. The system combines traditional credit data with selected alternative data sources, allowing it to model a broader view of borrower creditworthiness without excessive computational overhead. By embedding fairness metrics and mitigation strategies into the workflow, CredMill ensures compliance with ethical and regulatory standards while maintaining strong predictive capability. The final design delivers a practical, low-cost prototype capable of running on modest hardware or free cloud platforms, offering lenders a transparent, adaptable, and equitable tool for credit decision-making.

Introduction

Credit risk assessment lies at the heart of lending operations, influencing loan approvals, interest rate determination, and overall financial stability. For decades, banks and other lenders have relied on manual evaluations, fixed credit scoring formulas, and basic statistical methods to judge borrower reliability. These traditional approaches, while simple to implement, face mounting challenges: they are time-consuming, inconsistent across evaluators, and incapable of efficiently processing large, diverse datasets. In recent years, artificial intelligence (AI) and machine learning (ML) have shown promise in automating and refining credit assessments by uncovering complex, nonlinear relationships in financial and behavioral data. However, many existing AI solutions come with significant drawbacks: they operate as opaque "black boxes" that offer little interpretability, require high-end computational infrastructure or expensive APIs, and often fail to detect or mitigate demographic biases. Moreover, these solutions are typically built for large financial institutions, limiting their adaptability for smaller lenders or resource-constrained environments.

CredMill seeks to fill this gap by providing a transparent, fair, and computationally efficient credit scoring tool that leverages both traditional financial data and relevant alternative data sources. By combining explainable machine learning models with built-in fairness auditing and mitigation, the system not only improves prediction accuracy but also strengthens trust among stakeholders and ensures compliance with ethical standards. Designed to be trained and deployed on modest hardware or free cloud platforms, CredMill offers an accessible yet robust framework for modern credit risk assessment, especially in settings where transparency, fairness, and resource efficiency are critical.

Problem Statement

Traditional credit risk assessment methods rely heavily on manual evaluations, which are time-consuming, prone to human error, and inconsistent across evaluators. While modern AI-based solutions can improve predictive accuracy, they frequently operate as opaque “black box” models with minimal explanation provided to users or regulators, making them difficult to trust and audit. Many of these solutions also demand high-end computing resources or paid APIs for training and deployment, placing them out of reach for smaller institutions or student-level projects. Furthermore, existing models often fail to adequately detect or mitigate demographic biases in scoring, raising concerns over fairness and compliance. In addition, a large proportion of AI credit scoring tools are tailored for major financial institutions or limited to narrow data domains, restricting their broader applicability. The challenge, therefore, is to design and implement a resource-efficient, explainable, and fair AI-based credit risk assessment system that can effectively leverage both traditional financial data and alternative data sources, provide interpretable decision-making, and remain practical for small-to-medium-scale operational environments without requiring expensive infrastructure.

Proposed Solution

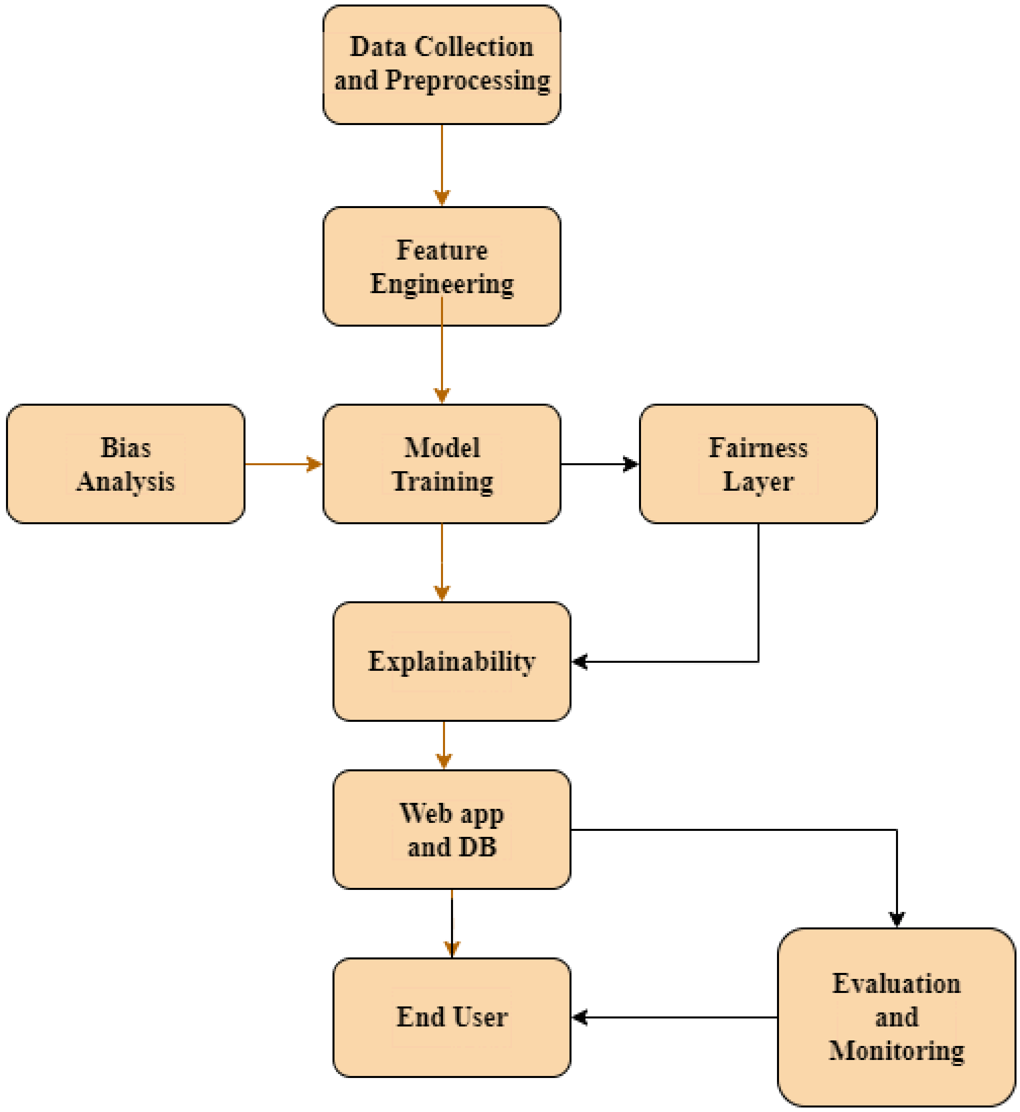
The proposed solution is a resource-efficient, explainable, and fair AI credit scoring system designed specifically for small-to-medium-scale operational environments; it will combine robust, CPU-friendly models with lightweight neural components only where sequential or textual signals add clear value, and it will integrate both traditional financial features (credit bureau scores, income, payment history) and curated alternative data.

To counter opacity, every predictive model will be paired with explainability layers—global explanations to describe how features drive overall model behavior and local explanations to justify individual decisions—using compute-aware XAI approaches (e.g., TreeSHAP sampling, approximate SHAP, and selective LIME analyses). Fairness will be an operational requirement: the pipeline will include pre-deployment bias audits, group-wise performance checks, and mitigation strategies such as reweighing or constrained optimization when disparate outcomes are detected. The user-facing outputs will be intentionally simple—probability of default, a short human-readable rationale showing the top contributing features, and recommended next-steps for loan officers—so the system supports transparency, regulatory review, and real-world decision-making without needing heavy infrastructure.

Methodology / Block Diagram

The methodology follows an iterative, data-centric approach that begins with careful curation of datasets and proceeds through reproducible preprocessing, model training, and interpretability checks; first, assemble a blend of curated public and institution-specific anonymized records, create interpretable features (debt-to-income, credit utilization, rolling payment behavior, counts of delinquencies) and optionally derive low-dimensional transaction summaries for alternative data, then apply robust preprocessing such as conservative imputation, clipped outlier handling, and encoding strategies that avoid leakage (target encoding applied within cross-validation folds). For modeling, start with interpretable baselines (logistic regression with class weighting and shallow decision trees) to set performance and interpretability baselines, then move to stronger but still resource-conscious learners (LightGBM/XGBoost with early stopping and constrained tree complexity), exploring small neural nets only if sequence/text signals justify them. Model selection will emphasize not just aggregate metrics but also calibration, stability across folds, and fairness metrics across demographic slices. For explainability, compute global feature importance and partial dependence plots on representative samples, and provide local case-level explanations using SHAP or LIME selectively; when compute is limited, use approximate or sampled explainability runs. The pipeline will be versioned (data slices, feature sets, model checkpoints) and validated with stratified or time-aware cross-validation to ensure robustness and to detect temporal drift.

Block Diagram:



Hardware , Software and tools Requirements

Hardware

- **Development environment**: A standard laptop (e.g., 8 GB RAM, Intel i5 / AMD Ryzen 5 or equivalent) — sufficient for development, testing, and lightweight model training.
- **Deployment**: A cloud environment or a small local VM/server capable of running a Flask backend and serving a lightweight UI.

Software / Libraries

Backend (API & Web Server)

- **Flask**: 3.1.1
  Optionally, Flask-related tooling (e.g., Flask-RESTX) can be added depending on API design.

Frontend (Web UI)

- **React** 19.1.1
- **Tailwind** 3.4.1

Machine Learning

- **scikit-learn**: 1.7.1
- **LightGBM**: 4.6.0
- **XGBoost**: 3.0.4

Deep Learning

- **TensorFlow** and **PyTorch**

Explainability & Fairness

- **SHAP**: 0.46.x
- **LIME**: 0.2.0.x
- **AIF360** (AI Fairness 360) and **Fairlearn**: latest stable releases (e.g., AIF360 ~0.6, Fairlearn ~0.10)

Database / Persistence

- **SQLite** (for simplified local storage), or **PostgreSQL** / **MySQL** (latest stable, e.g., MySQL 8.x), or **MongoDB** (latest 7.x release)

Other Tools

- **Git** (version control)
- **VS Code** (development IDE)
- **Postman** or **Insomnia** (API testing) — latest stable version

Proposed Evaluation Measures

- Statistical performance metrics will cover both overall and class-specific evaluation, including precision, recall (sensitivity), F1-score, AUC-ROC, precision-recall AUC, and calibration measures such as the Brier score. Class-wise reporting will ensure that minority-class performance remains visible and comparable.
- Business-aware measures will translate model outputs into expected monetary impacts through confusion-matrix cost modeling, profit curves, and the Expected Maximum Profit framework, enabling different operating points to be compared directly in financial terms.
- Interpretability and fairness will be assessed through the stability of top SHAP features across folds, as well as fairness diagnostics such as disparate impact ratios and group-wise ROC/AUC differences.
- Integrated evaluation will combine technical accuracy, business value, fairness, and interpretability into a multidimensional framework that connects statistical results with operational consequences, helping stakeholders choose models that align with both performance and ethical requirements.

Conclusion

In sum, CredMill aims to produce a pragmatic, resource-conscious credit scoring prototype that balances predictive strength with interpretability and fairness, specifically tailored for settings without enterprise compute budgets; by combining careful data curation, efficient modeling (favoring explainable ensembles), compute-aware XAI, and explicit fairness audits, the work will demonstrate how smaller lenders or teams can build responsible, effective credit risk tools that are auditable, adaptable, and aligned with operational and regulatory needs.

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